The galaxy-bias model of the lssty1xdesi paper

One model and one prior for DESI C_ℓ^{gg} in thin redshift bins

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1 Summary

The question. What does DESI galaxy clustering, measured as angular power spectra C_ℓ^{gg} in thin redshift bins, constrain on its own, with a bias model a referee accepts? Thin bins resolve the growth of structure along the line of sight only if the bias of neighboring bins is not free bin by bin; tying the bins too tightly instead turns assumptions about the bias into cosmology.

The model. The galaxy power spectrum is one-loop, with seven bias quantities per bin: the linear, quadratic and tidal biases, the counterterms α_0 and α_2, and the stochastic terms SN_0 and SN_2. Each of them is its group's value, which follows calibrations where they exist, plus, depending on the quantity, a free tilt in redshift across the group and a residual of the bin with a Gaussian prior correlated in redshift (Section 3). One principle sets the priors: slicing a wide DESI sample into thin bins adds measurements but no knowledge of the bias, so the group value and the tilt carry exactly the freedom of the wide sample, and the residual is what the slicing may add within it (Section 4).

The result. On the two thin layouts, $\mathrm{FoM}(w_0, w_a)$ of DESI C_ℓ^{gg} alone grows from 1.29 and 1.34 at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ to 10.9 and 13.9 at $0.2\,h\,\mathrm{Mpc}^{-1}$, below the 17.3 of LSST Y1 3x2pt alone; $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ stays below LSST Y1 as well (Section 5). Truths that hold one bias term at a full prior width in every bin move the fit by at most 0.59σ. Over random bias histories, the known drifts plus scatter drawn from the prior, the quoted error bars are honest at every cut and on every layout.

What the grouping is worth. Untying the bias bin by bin costs up to a factor of 5.5 in $\mathrm{FoM}(w_0, w_a)$ on the thin layouts at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$, while a per-bin prior that looks looser than the model gains spuriously from the slicing (Section 6).

2 Data and scale cut

The data are the DESI angular galaxy power spectra C_ℓ^{gg} in 53 bands of width $\Delta\ell = 10$ over $15 \le \ell \le 544$, with ℓ the multipole: the auto spectrum of every redshift bin and the cross spectra of neighboring and overlapping bins. The galaxies form six tracer groups g, each a wide sample selected as a whole: BGS (the Bright Galaxy Survey), LRG1–3 (luminous red galaxies) and ELG1–2 (emission-line galaxies). Four layouts slice them: $\Delta d_{\mathrm{com}} = 62$ Mpc, 76 bins of equal comoving width, and $\Delta z = 0.02$, 0.05 and 0.2, with 90, 36 and 10 bins of fixed redshift width. The first two are the thin layouts: a bin of comoving width $\Delta\chi$ resolves line-of-sight wavenumbers up to $k_\parallel \simeq \pi/\Delta\chi$, about $0.075\,h\,\mathrm{Mpc}^{-1}$ for the 62 Mpc bins, so thin bins keep much of the three-dimensional information [1]. Bin i has mean redshift $\bar{z}_i$, and $D(z)$ is the linear growth factor, $D(0) = 1$.

Non-Limber spectra. The Limber approximation evaluates the power spectrum only at the transverse wavenumber $(\ell + 1/2)/\chi$, with χ the comoving distance, and misses the line-of-sight modes that a thin bin with sharp edges passes. Every term of the model is therefore projected without it, the real-space power through a low-rank factorization of each bin's power spectrum and the stochastic terms as exact templates.

Parameters and figures of merit. Seven cosmological parameters vary with flat priors: $10^9 A_s$, with A_s the amplitude of the primordial curvature power spectrum; the spectral index n_s; the Hubble constant H_0; the baryon and matter densities today, Ω_b and Ω_m, in units of the critical density; and w_0 and w_a of the dark-energy equation of state $w_0 + w_a(1 - a)$, with a the scale factor. Seven Fingers-of-God velocity dispersions σ_v, which smooth the bins' redshift distributions, vary under Gaussian priors. With σ a parameter's marginalized one-sigma error, $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$ is the inverse area of the marginalized (w_0, w_a) ellipse, and $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ is defined the same way. The reference is LSST Y1 3x2pt alone [2].

The scale cut. Matter is less nonlinear at higher redshift, so bin i is cut at $k_{\max}/D(\bar{z}_i)$, and every $k_{\max}$ here is the $z = 0$ value. The cut is the non-Limber UV-fraction criterion of Saraivanov et al. [3], after DeRose and Chen [4]: each band power splits into C_ℓ^{IR}, from the power spectrum below the bin's cut wavenumber, and C_ℓ^{UV}, from the power above it, and a band is kept while

$$\frac{|C_\ell^{\mathrm{UV}}|}{|C_\ell^{\mathrm{UV}}| + |C_\ell^{\mathrm{IR}}|} \;<\; 0.05 \tag{1}$$

and within the shipped mask, which stops every bin at $0.176\,h\,\mathrm{Mpc}^{-1}$. The cut is built on linear-bias spectra. The forecasts scan $k_{\max} = 0.05, 0.075, \ldots, 0.2\,h\,\mathrm{Mpc}^{-1}$.

The covariance. The band-power covariance is Gaussian, computed at the one-loop fiducial (every bin's linear bias on its group's trend, b_2 and b_{s^2} on their calibrations, every other bias term zero) for 14,000 square degrees, with its inverse rescaled to the forecast's 12,300. The data vector sits at the same fiducial.

3 The bias model

The one-loop power spectrum. The galaxy power spectrum of bin i, in redshift space, is

$$\begin{aligned} P_{g,i}(k,\mu) \;&= b_{1,i}^2\, P_{\mathrm{mm}}(k) + P_{\mathrm{loop}}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) + \alpha_{0,i}\, k^2 P_{\mathrm{mm}}(k) \\ &\quad + \big(2\, b_{1,i} f\mu^2 + f^2\mu^4\big) P_{\mathrm{lin}}(k) + \big(b_{1,i} + f\mu^2\big) f\, \alpha_{2,i}\, \mu^2 k^2 P_{\mathrm{lin}}(k) \\ &\quad + \frac{1}{\bar{n}_i}\Big(\mathrm{SN}_{0,i} + \mathrm{SN}_{2,i}\, f_{\mathrm{sat}} \sigma_{\mathrm{sat}}^2\, k^2\mu^2\Big), \end{aligned} \tag{2}$$

with k the comoving wavenumber, μ the cosine of the angle to the line of sight, P_{mm} the nonlinear (halofit) matter power spectrum, P_{lin} the linear one, $f = d\ln D/d\ln a$ the growth rate and $\bar{n}_i$ the bin's number density. P_{loop} holds the one-loop terms of the quadratic bias b_2 and the tidal bias b_{s^2}, the responses of the galaxy density to the squared density contrast and to the squared tidal field [5, 6]; the third-order bias is zero. α_0 is the k^2 counterterm, which absorbs the sensitivity of the loops to small scales. The P_{lin} terms are linear (Kaiser) redshift-space distortions on each bin's Fingers-of-God-convolved redshift distribution, with no redshift-space one-loop terms, and α_2 is the line-of-sight counterterm in the convention of the DESI 2024 full-shape analysis [7], in

quantity	group value	tilt	residual
b_1	$B_g/D(\bar{z}_i)$; B_g flat	s_g, flat	ε_i, Eq. (7)
b_2	$b_2^{\rm L}(b_{1,i}) + \Delta_{2,g}$; $\Delta_{2,g}$ width 2	flat	none of its own
b_{s^2}	$-\frac{4}{7}(b_{1,i}-1) + \Delta_{s,g}$; $\Delta_{s,g}$ width 1	flat	none of its own
α_0, α_2	width $0.5/k_{\rm max}^2$	flat	none
SN_0	width 2	none	none
SN_2	width 5	none	none

Table 1. What each bias quantity carries in Eq. (3). A width is that of a Gaussian prior centered on zero, so the fiducial model sits on the calibrations with no counterterm and no stochastic term beyond Poisson; flat marks a flat prior. The counterterm width is the rule of the DESI 2024 full-shape analysis [9, 7], a correction of 0.5 at the cut, evaluated at each cut ($12.5\,h^{-2}\,\mathrm{Mpc}^2$ at $0.2\,h\,\mathrm{Mpc}^{-1}$). A group of one bin has no tilt.

which $\alpha_2 k^2$ is a fractional correction to the velocity part of the Kaiser power. Both counterterms are in $h^{-2}\,\mathrm{Mpc}^2$. SN_0 is the white noise beyond Poisson, in units of the Poisson power $1/\bar{n}_i$ that the covariance carries, and $\mathrm{SN}_2\,k^2\mu^2$ the stochastic term of the satellites' velocities, in units of $f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}_i$, with $f_{\rm sat}$ the satellite fraction and $\sigma_{\rm sat}$ the satellites' velocity dispersion in distance units: $(0.1, 7.8\,h^{-1}\mathrm{Mpc})$ for LRG and $(0.1, 2.9\,h^{-1}\mathrm{Mpc})$ for ELG [7], and the LRG values for BGS.

One form for every bias quantity. Every bias quantity X of Eq. (2), in bin i of group g, is

$$X_i \;=\; X_g(\bar{z}_i) \;+\; t_{X,g}\,(\bar{z}_i - \bar{z}_g) \;+\; \varepsilon_{X,i}, \tag{3}$$

with $X_g(\bar{z}_i)$ the group value of X, which includes the calibration trend where X has one, $t_{X,g}$ a tilt in redshift across the group, $\bar{z}_g$ the mean redshift of the group's bins, and $\varepsilon_{X,i}$ a residual of the bin. The group value and the tilt are the freedom of the wide sample; the residual is what thin slicing may add within it (Section 4). Table 1 lists what each quantity carries.

For the linear bias the three parts enter as fractions of the group value $B_g/D(\bar{z}_i)$, which keeps the clustering amplitude $b_1 D$ constant across the group:

$$b_{1,i} \;=\; \frac{B_g}{D(\bar{z}_i)}\,\big(1+\delta_i\big), \qquad \delta_i \;=\; s_g\,(\bar{z}_i - \bar{z}_g) + \varepsilon_i, \tag{4}$$

with B_g the group amplitude, s_g the tilt, ε_i the residual and δ_i the fractional deviation from the group value. The group values of the quadratic and tidal biases are calibrations evaluated at the bin's own linear bias, plus a group offset:

$$b_{2,i} = b_2^{\rm L}(b_{1,i}) + \Delta_{2,g} + t_{2,g}\,(\bar{z}_i - \bar{z}_g), \qquad b_{s^2,i} = -\tfrac{4}{7}\,\big(b_{1,i}-1\big) + \Delta_{s,g} + t_{s,g}\,(\bar{z}_i - \bar{z}_g), \tag{5}$$

with $b_2^{\rm L}(b_1) = 0.412 - 2.143\,b_1 + 0.929\,b_1^2 + 0.008\,b_1^3$ the separate-universe fit of Lazeyras et al. [8] and the second relation the co-evolution of a population formed with local bias [6]. Only the linear bias carries a residual of its own: the quadratic and tidal biases receive each bin's residual through $b_{1,i}$ in their calibrations, and the counterterms and stochastic terms have none.

Why SN_2 matters on thin bins. A constant power N projects exactly as $N\int d\chi\, W_i W_j/\chi^2$ onto bins i and j, with $W_i(\chi)$ the Fingers-of-God-convolved kernel of bin i; bins of two different samples

share no stochastic term. A power proportional to $k^2\mu^2 = k_\parallel^2$ is that of the line-of-sight derivative of a white noise, so SN_2 projects with each kernel replaced by its derivative, $W_i' = dW_i/d\chi$:

$$C_\ell^{ij}\big|_{\mathrm{SN}_2} \;=\; U_{ij} \int d\chi\, \frac{W_i'(\chi)\, W_j'(\chi)}{\chi^2}, \tag{6}$$

with U_{ij} the mean of the two bins' amplitudes $\mathrm{SN}_2 f_{\mathrm{sat}} \sigma_{\mathrm{sat}}^2/\bar{n}$. A thin bin's W_i' peaks at its two edges with opposite signs, and adjacent bins share an edge, where one kernel rises as the other falls: the projected SN_2 of neighbors is anticorrelated. Normalized by the two auto terms, the adjacent term has a median of -0.50 over the 73 adjacent pairs of one sample on $\Delta d_{\mathrm{com}} = 62$ Mpc, against $+0.05$ for SN_0. No white noise, per group or per bin, mimics this pattern, and a fit without SN_2 is biased on the thin layouts already at $k_{\max} = 0.075\, h\,\mathrm{Mpc}^{-1}$ (`notes/higher_order_bias`).

4 The prior: the freedom of the wide sample

The principle. DESI selects each group as one wide sample, and what is known about its bias, from calibrations to the priors of full-shape analyses, refers to that sample. Slicing it into thin bins adds measurements of the clustering, not knowledge of the bias. A thin-bin prior is therefore fair only if it leaves each group the freedom of its wide sample, whatever the slicing. In Eq. (3) that freedom is the group value and the tilt, whose priors are set for the wide sample; the residual is what the slicing may add within it, local features of the size the selection can produce, and its prior must not tighten as the bins get thinner.

The linear bias. The level B_g and the tilt s_g are free, and the residual has a Gaussian prior, zero across groups and correlated within one,

$$\big\langle \varepsilon_i\, \varepsilon_j \big\rangle \;=\; \sigma_g^2\, \exp\!\left(- \frac{|\bar{z}_i - \bar{z}_j|}{L} \right), \tag{7}$$

with σ_g the width of one bin's residual and L the correlation length in redshift. The kernel defines a random process $\varepsilon(z)$ along each group before any binning, which a layout only samples at its bin redshifts: the prior on a coarse set of redshifts is the marginal of the prior on a finer one, so slicing finer adds no prior knowledge. Two neighbors at separation Δz differ by an rms $\sigma_g[2(1-e^{-\Delta z/L})]^{1/2}$. The prior charges only the residual around each group's best straight line (Section 7), so a group of one or two bins, as on $\Delta z = 0.2$, has a free bias in every bin.

No measurement resolves the bias of these samples below $\Delta z \approx 0.1$, so the sizes come from what selects them (`notes/correlated_bias_prior_v2`). Coherent changes move a whole group along a nearly straight line, which the free tilt absorbs: a conserved population, whose bias obeys $b_1 - 1 \propto 1/D$ [10], departs from B_g/D by about 1% per $\Delta z = 0.05$, and the $r < 19.5$ flux limit of BGS [11] drifts its bias by 22% across the group. Local changes, such as the 4000 Å break crossing a filter edge over $\Delta z \approx 0.03$–0.05, move bins independently. Hence $\sigma_g = 3\%$ in every group, above the largest derived departure from a line (the 2.4% rms that the BGS flux limit leaves), and $L = 0.05$, the scale of a filter-edge crossing; neighbors at $\Delta z = 0.02$ then differ by 2.4% rms. The draws of Figure 1 keep the line of the known drift and add local features of a few percent.

The higher-order terms. The calibrations of b_2 and b_{s^2} describe halo populations; a wide sample adds how far it sits from them. Each group therefore gets one offset per family, with the widths of Table 1: twice what the known imperfection of the calibrations requires (the scatter of

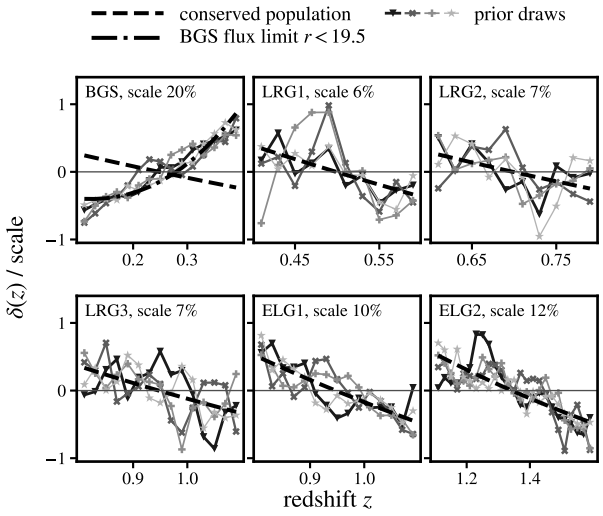


Figure 1. Four draws of the prior on the $\Delta z = 0.02$ bins, one panel per group, each panel divided by the round scale written in it. Thick black dashed: the conserved-population drift; thick black dash-dot (BGS only): the $r < 19.5$ flux-limit drift. Thin gray lines: the draws of $\delta(z)$, each with its own shade, line width and marker. Every draw keeps the best straight line through its group's reference drift, the flux limit in BGS and the conserved population elsewhere, and differs from it by a residual drawn with $\sigma_g = 3\%$ and $L = 0.05$. The level of every curve is removed, since B_g is free.

the Lazeyras fit and the 30% spread between co-evolution and excursion-set tidal biases), so that the calibrations act as prior centers where the loops reach 15% of the galaxy power. A calibration of the wide sample says nothing about how the offset changes inside it, so the tilt is free; it also absorbs exactly a calibration that drifts linearly across a group. The counterterms follow the same pattern, since their scale follows the host halos of the selection. The stochastic amplitudes do not tilt: their Poisson part is known bin by bin.

5 Forecast

Figures of merit. Figure 2 draws both FoMs against $k_{\rm max}$. FoM(w_0, w_a) is 1.29, 1.34, 0.227 and 0.0495 at $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$ and 10.9, 13.9, 2.64 and 0.289 at $0.2\,h\,{\rm Mpc}^{-1}$, in layout order ($\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2); FoM$(\Omega_m, 10^9 A_s)$ at $0.2\,h\,{\rm Mpc}^{-1}$ is 70.8, 81.3, 23.1 and 2.98. The thin layouts carry the information: from $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$ on, their FoM(w_0, w_a) is several times that of $\Delta z = 0.05$ and tens of times that of $\Delta z = 0.2$. LSST Y1 3x2pt alone gives 17.3 and 215; DESI C_ℓ^{gg} alone stays below it in both FoMs at every cut.

A stress test. Figure 3 confronts the model with truths at the edge of its priors, each holding one term at a full prior width with the same value in every bin: $\alpha_2 = \pm 0.5/k_{\rm max}^2$, ${\rm SN}_2 = \pm 5$, or $(\Delta_2, \Delta_s) = \pm(2, 1)$; a fourth, the calibration drift, is absorbed exactly by the tilts. A fit's shift is its linear response (Section 7), in units of its own marginalized ellipse for (w_0, w_a) and of σ for Ω_m and A_s. From $k_{\rm max} = 0.1\,h\,{\rm Mpc}^{-1}$ on, the largest shift on the three thinner layouts lies between 0.32σ and 0.59σ, set mostly by α_2; on $\Delta z = 0.2$, whose wide bins barely see α_2, the offsets set it, at most 0.23σ at any cut.

The error bars are honest. How well a fit's quoted error bars cover where the fit actually lands is measured by $\eta = \sqrt{\langle \chi^2 \rangle / n_{\rm dof}}$, with $\chi^2 = \Delta p^{\mathsf{T}} \mathsf{C}_p^{-1} \Delta p$, Δp the recovered minus the true parameters, C_p the fit's marginalized covariance, $n_{\rm dof} = 2$ for (w_0, w_a) and 1 for Ω_m or A_s, and $\langle \cdot \rangle$ the average over the truths and the data noise; $\eta = 1$ for honest error bars. Over random bias histories, the known drifts plus a residual drawn from the prior, $\eta \leq 1.023$ at every cut and on

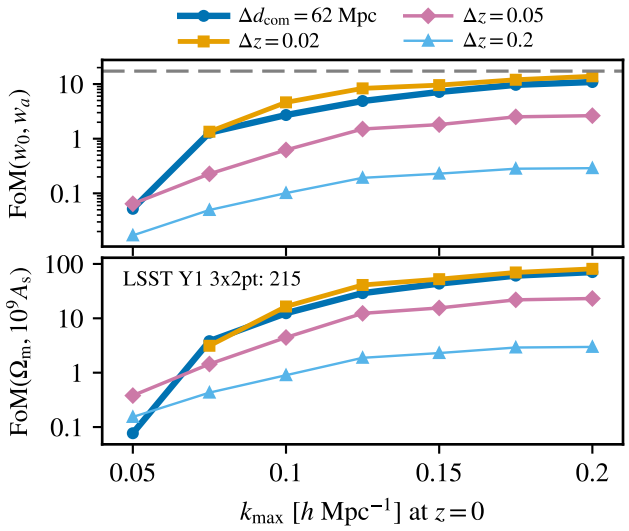


Figure 2. FoM(w_0, w_a) (top) and FoM$(\Omega_m, 10^9 A_s)$ (bottom) of DESI C_ℓ^{gg} alone against $k_{\rm max}$. Color, line width and marker: the layouts. Gray dashed line (top) and text (bottom): LSST Y1 3x2pt alone. On $\Delta z = 0.02$ the cut at 0.05 keeps no band. $k_{\rm max}$ at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

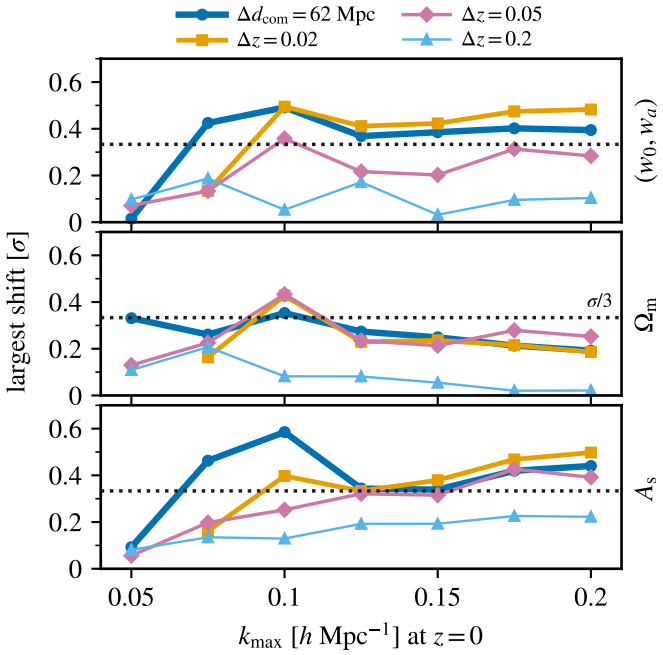


Figure 3. The largest shift of a fit over four truths, each the same in every bin: α_2, SN_2 and the offsets (Δ_2, Δ_s) at plus or minus one prior width, and the calibration drift (Δ_2 and Δ_s changing linearly by 1 and 0.5 across each group), which the tilts absorb exactly. Top: (w_0, w_a), in units of the fit's marginalized ellipse; middle and bottom: Ω_m and A_s, in units of their marginalized σ; the three panels share one linear axis from 0. Color, line width and marker: the layouts; dotted line, labeled in place: $\sigma/3$. On $\Delta z = 0.02$ the cut at 0.05 keeps no band. $k_{\rm max}$ at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

every layout (at most 1.0125 for (w_0, w_a), 1.0232 for Ω_m and 1.0034 for A_s), and the 1σ region covers the truth in at least 67.1% of universes, against 68.3% for honest error bars.

6 What the grouping is worth

A thin-bin prior can look loose and still add knowledge the wide sample does not have. Figure 4 compares the model with four versions that untie the bias bin by bin, with $b_{1,i}$ free and one value per bin of every other quantity; w is the width the model gives a group parameter and N the number of bins of the group.

- **Width w per bin** looks looser than the model, yet its FoM lies above the model's on most

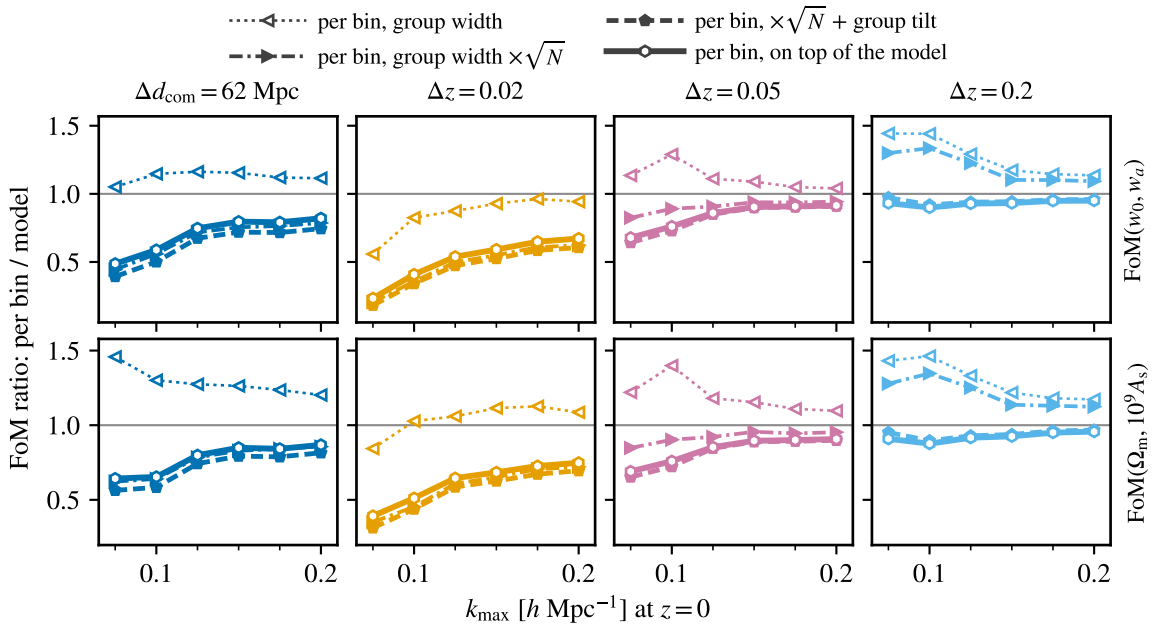


Figure 4. The FoM of four versions with the bias untied bin by bin, divided by that of the model: FoM(w_0, w_a) (top row) and FoM$(\Omega_m, 10^9 A_s)$ (bottom row), one column per layout. Every version frees $b_{1,i}$ in each bin and gives Δ_2, Δ_s, α_0, α_2, SN_0 and SN_2 one value per bin; the group width is the width the model gives the group parameter, and N the number of bins of the group. Dotted, open left triangles: the group width; dash-dot, filled right triangles: the group width times $\sqrt{N}$; dashed, filled pentagons: the same plus a free tilt per group; solid, open hexagons: the group width on top of the model's offsets and tilts. The line width grows in that order. Thin gray line: a ratio of 1. The cuts start at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$: at $0.05\,h\,\mathrm{Mpc}^{-1}$ the model's own FoM(w_0, w_a) is below 0.1. $k_{\max}$ at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

layouts, by up to a factor of about 1.5: N independent values of width w pin their average to $w/\sqrt{N}$, so the group's common shift, which the wide sample knows only to w, becomes better known the finer the slicing. The gain is spurious.

- **Width $w\sqrt{N}$ per bin** gives the common shift back its width w, but on $\Delta z = 0.2$, whose groups hold one or two bins, it still fixes the trend that the model leaves free.
- **Width $w\sqrt{N}$ plus a free tilt per group** restores the trend too. Its prior is at least as loose as the model's in every direction (a variance Nw^2 per bin against w^2 for a group's common shift), so it can only lose FoM: it measures what untying every bin costs.
- **Width w on top of the model** gives the same cost from the other side.

The last two agree roughly: untying costs a factor of 2.0 to 5.5 in FoM(w_0, w_a) on the thin layouts at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$, 1.1 to 1.6 on the three thinner layouts at $0.2\,h\,\mathrm{Mpc}^{-1}$, and at most 1.1 on $\Delta z = 0.2$; at $0.05\,h\,\mathrm{Mpc}^{-1}$, where the model's own FoM(w_0, w_a) is below 0.1, it costs more.

7 Method

Every forecast is a Fisher forecast at the fiducial point. With $\mathbf{m}$ the band powers, C their covariance and D the derivatives of $\mathbf{m}$ with respect to the parameters (sans-serif, unlike the growth factor D),

a fit's Fisher matrix on the band powers its cut keeps is $F = \mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{D} + F_{\rm prior}$, and F^{-1} gives the errors and the FoMs. The derivatives with respect to the cosmology, the σ_v and four operators in every bin (b_1, b_2, b_{s^2}, α_0) are five-point stencils of the full likelihood, CAMB included, which the chain rule combines into the column of every parameter of the model; the terms linear in their amplitude, α_2, SN_0 and SN_2, enter as exact templates. $F_{\rm prior}$ holds the Gaussian priors and, for each group and each quantity that carries a residual (Table 1), the inverse of the kernel of Eq. (7) projected orthogonally to the level and the line, so that it charges only the residual around the best straight line.

A truth that moves the band powers by $\delta\mathbf{m}$ moves the fit by $\delta\mathbf{p} = F^{-1}\mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\,\delta\mathbf{m}$, exactly for the α_2 and SN_2 truths, whose terms are linear, and to first order otherwise. The response is linear in the bias history, so $\langle\chi^2\rangle$ is the exact average over infinitely many histories and data realizations, in closed form.

8 Where the details are

- `notes/correlated_bias_prior_v2`: the linear-bias prior and its physics, the projected precision, the random bias histories and the error-bar test, and the comparisons of priors and models.
- `notes/higher_order_bias`: the one-loop model, its calibrations, the size and exact projection of every term, and its validation.
- `notes/linear_bias_kmax`: the reach of linear bias, which motivates the one-loop terms.
- `notes/fisher_machinery`: the Fisher derivatives and the assembly of every Fisher matrix, shift and $\langle\chi^2\rangle$.

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